A bound for t-$(v,k,1)$ packings with $t \geq 3$

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Notation. Throughout, $v \geq k \geq t \geq 2$ are integers, and V is a set of v points. Put $Q = \binom{k}{t}$ and $b = \binom{k-1}{t-1}$, so that $kb = tQ$.

A t-$(v,k,1)$ *packing* is a collection P of k-subsets of V (*blocks*) such that every t-subset of V lies in at most one block. Two blocks then meet in at most $t-1$ points, that is, they are at Hamming distance at least $2(k-t+1)$. So the maximum of $|P|$ is $A(v, 2(k-t+1), k)$, where $A(n,d,w)$ is the maximum size of a binary code of length n, constant weight w and minimum distance at least d.

The *leave* of P is the t-uniform hypergraph L on V whose edges are the t-subsets of V that lie in no block.

Theorem. *Let $t \geq 3$. Suppose that $\binom{k-s}{t-s}$ divides $\binom{v-s}{t-s}$ for $s = 1, \ldots, t-1$, and that $\binom{k}{t}$ does not divide $\binom{v}{t}$. Then*

$$|P| \;\leq\; \left\lfloor \binom{v}{t} \Big/ \binom{k}{t} \right\rfloor - 2.$$

Examples. Let $t = 4$ and $k = 7$, so that $\binom{6}{3} = 20$, $\binom{5}{2} = 10$, $\binom{4}{1} = 4$ and $Q = \binom{7}{4} = 35$.

For $v = 27$: $\binom{26}{3} = 2600 = 130 \cdot 20$, $\binom{25}{2} = 300 = 30 \cdot 10$, $\binom{24}{1} = 24 = 6 \cdot 4$, and $\binom{27}{4} = 17550 = 501 \cdot 35 + 15$. Hence $A(27,8,7) \leq 499$.

For $v = 47$: $\binom{46}{3} = 15180 = 759 \cdot 20$, $\binom{45}{2} = 990 = 99 \cdot 10$, $\binom{44}{1} = 44 = 11 \cdot 4$, and $\binom{47}{4} = 178365 = 5096 \cdot 35 + 5$. Hence $A(47,8,7) \leq 5094$.

The reason is this. In the leave, the number of edges on a set of s points, $1 \leq s \leq t-1$, is divisible as it is in a complete hypergraph on k points. A hypergraph with this property is empty, or complete on k points, or has at least $2Q$ edges.

1. The degrees of the leave. For a set S of points, $\deg(S)$ is the number of edges containing S. Let $1 \leq |S| = s \leq t-1$. There are $\binom{v-s}{t-s}$ t-subsets of V that contain S. A block containing S contains $\binom{k-s}{t-s}$ of them, and no t-subset lies in two blocks. So in the leave

$$\deg(S) = \binom{v-s}{t-s} - m\binom{k-s}{t-s},$$

where m is the number of blocks containing S. By assumption this is divisible by $\binom{k-s}{t-s}$. So the leave has the property

(D) $\deg(S)$ is divisible by $\binom{k-s}{t-s}$ whenever $1 \leq |S| = s \leq t-1$.

Each block contains Q t-subsets, and no t-subset lies in two blocks. So the leave has $e = \binom{v}{t} - |P|\,Q$ edges.

2. Hypergraphs with property (D).

Lemma 1. *Let $k \ge t \ge 2$, and let L be a t-uniform hypergraph without repeated edges, with e edges, satisfying* (D)*. Then every point lies on at most te/k edges. If $e > 0$ then $e \ge Q$, and if $e = Q$ then L is the complete t-uniform hypergraph on k points (and isolated points).*

Proof. Write $d(x) = \deg(\{x\})$. By (D) it is a multiple of b. The sum of $d(x)$ over all points is te. We show $d(x) \le te/k$ by induction on t.

The case $t = 2$. L is a graph in which every degree is a multiple of $k-1$. Let x have degree $D > 0$. Each neighbour of x has positive degree, hence degree at least $k-1$. The degrees of x and of its D neighbours add up to at least $D + D(k-1) = kD$, and to at most $2e$. So $D \le 2e/k$.

The case $t \ge 3$. For a point u, let L_u be the $(t-1)$-uniform hypergraph whose edges are the sets $E \setminus \{u\}$, for the edges E of L on u. It has $d(u)$ edges, none repeated. If S does not contain u and $1 \le |S| = s \le t-2$, then S lies on $\deg(S \cup \{u\})$ edges of L_u, and by (D) this number is divisible by $\binom{k-s-1}{t-s-1}$. So L_u satisfies (D) with $t-1$, $k-1$ instead of t, k. By induction, a point y other than u lies on at most $(t-1)\,d(u)/(k-1)$ edges of L_u:

$$\deg(\{u, y\}) \le \frac{(t-1)\,d(u)}{k-1}. \tag{1}$$

Fix y and add (1) over all u other than y. On the left, each edge on y is counted $t-1$ times, so the sum is $(t-1)\,d(y)$. On the right, the sum of $d(u)$ over u other than y is $te - d(y)$. Hence $(k-1)\,d(y) \le te - d(y)$, that is, $d(y) \le te/k$.

The last two statements. If $e > 0$, some point has $d(x) > 0$, hence $d(x) \ge b$, and $e \ge kb/t = Q$. If $e = Q$, then $d(x) \le tQ/k = b$ for all x, so every point on an edge has $d(x) = b$, and there are $te/b = k$ such points. The Q edges are distinct t-subsets of these k points, and there are only $\binom{k}{t} = Q$ such subsets. So every one is an edge. □

Lemma 2. *Let $t \ge 3$, and let L be as in Lemma 1. Then $e = 0$, $e = Q$, or $e \ge 2Q$.*

Proof. Suppose $0 < e < 2Q$. For a point x on an edge, let $N(x)$ be the set of the other points that lie on an edge with x.

(a) Let x be on an edge. By Lemma 1, $d(x) \le te/k < 2b$, so $d(x) = b$. Then L_x has $b = \binom{k-1}{t-1}$ edges and satisfies (D) with $t-1 \ge 2$ and $k-1$. By the last part of Lemma 1, L_x is the complete $(t-1)$-uniform hypergraph on $k-1$ points. These points are $N(x)$. So $|N(x)| = k-1$, and every $(t-1)$-subset of $N(x)$, with x added, is an edge.

(b) Let x be on an edge and $W = N(x) \cup \{x\}$. Then $N(y) = W \setminus \{y\}$ for every $y \in W$. For $y = x$ this is the definition of W. Let $y \in N(x)$. Every other point z of W is on an edge with y: if $z = x$, by the definition of $N(x)$; if $z \in N(x)$, take a $(t-1)$-subset of $N(x)$ containing y and z (possible, as $2 \le t-1 \le k-1$) and add x. So $N(y)$ contains $W \setminus \{y\}$. By (a) both have $k-1$ points, so they are equal.

(c) By (b), an edge on a point y of W lies inside W, since its other points are in $N(y)$. And every t-subset T of W is an edge: take $y \in T$; then $T \setminus \{y\}$ is a $(t-1)$-subset of $N(y)$, and by (a) T is an edge. So W carries a complete hypergraph with Q edges, and no other edge meets W.

(d) If some point x' on an edge is outside W, form $W' = N(x') \cup \{x'\}$ in the same way. If W' met W, the complete hypergraph on W' would have an edge containing x' and a point of W, against (c). So W' is disjoint from W. Continuing, the edges of L fall into complete hypergraphs on disjoint k-sets, each with Q edges. Since $0 < e < 2Q$ there is exactly one, and $e = Q$. □

Proof of the Theorem. By Section 1, the leave satisfies (D) and has $e = \binom{v}{t} - |P|\,Q$ edges. Let r be the remainder of $\binom{v}{t}$ on division by Q; by assumption $1 \le r \le Q-1$. Then $e = aQ + r$ for an integer a, so e is neither 0 nor Q. By Lemma 2, $e \ge 2Q$, hence $a \ge 2$ and $e \ge 2Q + r$. Therefore

$$|P| = \frac{\binom{v}{t} - e}{Q} \;\le\; \frac{\binom{v}{t} - r}{Q} - 2 = \left\lfloor \binom{v}{t} \Big/ Q \right\rfloor - 2. \qquad \square$$

Remarks. Lemma 2 fails for $t = 2$: the 4-cycle has all degrees divisible by 2 (here $k = 3$) and has 4 edges, between $Q = 3$ and $2Q = 6$.

For $t = 3$, Javadi, Poorhadi and Fallah [2] prove the bound with -3 instead of -2 under the same conditions.

The argument is close to pages 89–91 of [1].

References

[1] A. E. Brouwer, Packing and covering of $\binom{k}{t}$-sets, in: A. Schrijver (ed.), *Packing and Covering in Combinatorics*, Math. Centre Tracts 106, Amsterdam, 1979, pp. 89–97.

[2] R. Javadi, E. Poorhadi and F. Fallah, Packing cliques in 3-uniform hypergraphs, *J. Combin. Des.* **28** (2020) 580–603; Theorem 13 in arXiv:1905.10807.